George Panagopoulos

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Experience

- **Amazon**, Applied Scientist, Luxembourg (04.2022 - 06.2023)
  - Machine learning for routing (internship), operations research (FTE)
- **Ecole Polytechnique Informatics Laboratory (LIX)**, RA/TA, Paris, France (06.2018 - 03.2022)
  - **Graph and reinforcement learning** for combinatorial optimization and forecasting
  - **Data mining** from big bibliographical networks, legal documents, medical ontologies
- **University of Houston, Computational Physiology Lab**, RA/TA, Houston, TX (08.2016 - 05.2018)
  - **Forecasting** driving anomalies using physiological signals from wearable sensors
  - **Statistical analysis and inference** on behavioral data
- **National Center for Scientific Research Demokritos**, Research Assistant, Athens, Greece (08.2014 - 07.2016)
  - **Brain computer interface** experiments and tutorials for university and high-school students
  - **Natural language processing** modeling computational creativity in multi-lingual essays

Education

- **Ph.D. in Computer Science**, Ecole Polytechnique, Paris, France (01.2019 - 03.2022)
  - Thesis Title: *"Learning Influence Representations: Methods and Applications"*
  - Supervisors: Prof. Michalis Vazirgiannis , Asst Prof. Fragkiskos Malliaros
- **Ms. in Computer Science**, University of Houston, Houston, TX (08.2016 - 05.2018)
  - Thesis Title: *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"*
  - Supervisor: Prof. Ioannis Pavlidis
- **Bsc. in Informatics and Telematics**, Harokopio University of Athens, Greece (09.2010 - 07.2014)
  - Thesis Title: *"Detecting Rising Stars in Dynamic Collaborative Networks"*
  - Supervisor: Prof. Iraklis Varlamis

Selected Publications

1. G. Panagopoulos, N. Tziortziotis, F. Malliaros, M. Vazirgiannis. *"Learning Graph Representations for Influence Maximization"* Analysis of Socian Networks and Mining (**ASONAM**), (2023) [TO APPEAR]
2. C. Kosma, G. Nikolentzos, G. Panagopoulos, J.M. Steyaert, M. Vazirgiannis. "Neural Ordinary Differential Equations for Modeling Epidemic Spreading", in **Transactions on Machine Learning Research** [TO APPEAR]
3. B. Rozemberczki , P. Scherer, Y. He, G. Panagopoulos, M. Astefanoaei, O. Kiss, F. Beres, N. Collignon, R. Sarkar. *"PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models"*. in Conference on Information and Knowledge Management (**CIKM resource track**), 2021
4. G. Panagopoulos, G. Nikolentzos, M. Vazirgiannis.*"Transfer Graph Neural Networks for Pandemic Forecasting"*, in AAAI Conference on Artificial intelligence (**AAAI**), 2021
5. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Multi-task Learning for Influence Estimation and Maximization"*, in **IEEE Transactions of Knowledge and Data Engineering**, 2020, 10.1109/TKDE.2020.3040028
6. G. Panagopoulos, H. Jalalzai *"Graph Neural Networks with Extreme Nodes Discrimination"*, in Deep Learning on Graphs: Methods and Applications, (**DLG@KDD**) , 2020
7. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Influence Maximization using Influence and Susceptibility Embeddings"*, in AAAI International Conference on Web and Social Media (**ICWSM**), 2020
8. G. Panagopoulos, I. Pavlidis. *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"* **IEEE Transactions of Intelligent Transportation Systems** (2019)

Technical Skills

- Coding: Python (advanced), Java (intermediate), Scala, C++ (basic)
- Machine Learning: PyTorch, PyG (advanced), JAX, Tensorflow (intermediate)
- Statistics: R (advanced), Matlab (intermediate), Julia (basic)
- Databases: MySQL, PostgreSQL (advanced), Spark, Hadoop (basic)
- Optimization: XPRESS (intermediate), CVXPY (basic)